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・指数型① $a_{n+1} = pa_n + q\cdot r^n$

両辺 r^{n+1} でわって

$$\underbrace{\frac{a_{n+1}}{r^{n+1}}}_{b_{n+1}} = \frac{p}{r}\cdot\underbrace{\frac{a_n}{r^n}}_{b_n} + \frac{q}{r}$$

と変形して㋑特殊解型へ

6A $a_{n+1} = 2a_n + 4\cdot 3^n \cdots ①$, $a_1 = 1$

(解法1)

①の両辺を 3^{n+1} でわって

$$\frac{a_{n+1}}{3^{n+1}} = \frac{2}{3}\cdot\frac{a_n}{3^n} + \frac{4}{3}$$

$\frac{a_n}{3^n} = b_n$ とおくと ← $\frac{a_{n+1}}{3^{n+1}} = b_{n+1}$

$$\begin{cases} b_{n+1} = \frac{2}{3}b_n + \frac{4}{3} \\ b_1 = \frac{a_1}{3^1} = \frac{1}{3} \end{cases}$$

← ㋑特殊解型に帰着

よって

$b_{n+1} - 4 = \frac{2}{3}(b_n - 4)$ ← $x = \frac{2}{3}x + \frac{4}{3}$ $\therefore x = 4$

($\{b_n - 4\}$ は㊥ $b_1 - 4$ ㊣ $\frac{2}{3}$ の等比数列)

$$b_n - 4 = (b_1 - 4)\cdot\left(\frac{2}{3}\right)^{n-1}$$

$$= -\frac{11}{3}\cdot\left(\frac{2}{3}\right)^{n-1}$$

$$\therefore b_n = 4 - \frac{11}{3}\cdot\left(\frac{2}{3}\right)^{n-1}$$

ここで, $\frac{a_n}{3^n} = b_n$ より

$$a_n = 3^n\cdot b_n$$

$$= 3^n\left\{4 - \frac{11}{3}\cdot\left(\frac{2}{3}\right)^{n-1}\right\}$$

$$= 4\cdot 3^n - 11\cdot 2^{n-1} \,//$$

6A $a_{n+1} = 2a_n + 4\cdot 3^n \cdots ①$, $a_1 = 1$

(解法2)

$\boxed{a_{n+1} + A\cdot 3^{n+1}} = 2\boxed{(a_n + A\cdot 3^n)}$ ← 幸せな形

指数部

nをn+1にした式

$$a_{n+1} + 3A\cdot 3^n = 2a_n + 2A\cdot 3^n$$

$$a_{n+1} = 2a_n - A\cdot 3^n$$

①と係数比較して

$$A = -4$$

$$a_{n+1} - 4\cdot 3^{n+1} = 2(a_n - 4\cdot 3^n)$$

($\{a_n - 4\cdot 3^n\}$ は㊥ $a_1 - 4\cdot 3^1$ ㊣ 2 の等比数列)

$$a_n - 4\cdot 3^n = (a_1 - 4\cdot 3^1)\cdot 2^{n-1}$$

$$= -11\cdot 2^{n-1}$$

$$\therefore a_n = 4\cdot 3^n - 11\cdot 2^{n-1} \,//$$